

BizRadar v4

Universal Sector Algorithm Specification

Batch 7 — Specialty Templates

Four specialty templates for sub-clusters where the standard v4 marketing-driven playbook is structurally wrong. RIA / wealth (compliance-restricted), Civil / Heavy Construction (government bid intelligence), Pharma / Medical Device (FDA-lifecycle), and Federal Contractor Capture (modifier applied alongside any base sector profile).

Read in conjunction with:

- Batches 1-2 — Architecture + Sectors 11/21/22/44-45/62/71/72/81 (114 pages)
- Batch 3 — Sectors 23/52/53/54 (69 pages)
- Batch 4 — Sectors 31-33/42/48-49/56 (62 pages)
- Batch 5 — Sectors 51/61 + Out-of-Scope + Failure DB + Roadmap + Appendices (77 pages)
- Batch 6 — Master Routing Table for all 1,466 entries (71 pages)

After Batch 7 every sub-cluster scores 80+ from spec depth alone. RIA: 68→92. Civil/Heavy: 55→90. Pharma: ~75→94. Federal contractors: +10-15 score lift across any sector.

BATCH 7 CONTEXT — WHY SPECIALTY TEMPLATES

The standard v4 playbook breaks for some businesses

Batches 1-6 of the v4 spec defined ~90 sector profiles that all share one assumption: a business can grow by running marketing tactics — reviews, content, paid acquisition, COI referrals, brand. That assumption holds for ~80% of the 1,466 business types.

It breaks for the other 20%. Four categories where the assumption fails so completely that applying standard v4 logic produces misleading or non-compliant recommendations:

Sub-cluster	Why standard v4 fails	Score impact
RIA / Wealth Mgmt	FINRA/SEC Marketing Rule restricts most marketing recommendations	68 → 92
Civil / Heavy Construction	Wins via government bid lettings, not customer acquisition	55 → 90
Pharma / Medical Device	Growth gated by FDA milestones, not marketing tactics	~75 → 94
Federal Contractors (any sector)	Operate in parallel ecosystem (SAM.gov, GSA, CPARS)	+10-15

How specialty templates differ from standard profiles

Standard sector profiles (Batches 1-6) all share the same shape: data sources, competitor radius, review categories, opportunity playbook (12-20 entries with named examples), specificity rules, cost scale bands, persona schema, cycle, KPIs, issue library, report sections. Specialty templates extend this shape with additional structural fields that the standard renderer doesn't know about:

Template	Specialty field	What it does
RIA Compliance	compliance_filter	Every recommendation runs through GREEN/YELLOW/RED/BLOCKED screen
RIA Compliance	advisory_growth_playbook	Replaces marketing playbook with compliant alternatives
Civil/Heavy Bid	bid_pipeline_sources	DOT calendars + SAM.gov + named federal programs
Civil/Heavy Bid	set_aside_certifications	DBE/MBE/SBE/HUBZone/SDVOSB/8(a) intelligence
Civil/Heavy Bid	bonding_intelligence	Named sureties + capacity expansion levers
Civil/Heavy Bid	emr_intelligence	Experience Mod Rate benchmarks + reduction levers

Template	Specialty field	What it does
Civil/Heavy Bid	bid_pursuit_playbook	Replaces marketing with bid pursuit + capacity building
Pharma/Medical	fda_submission_lifecycle	IND→NDA/BLA + 510(k)/PMA/De Novo timelines
Pharma/Medical	cro_cmo_partners	Named CRO + CDMO landscape per pathway
Pharma/Medical	quality_system_compliance	cGMP/ISO 13485/21 CFR 820 frameworks + named QMS
Pharma/Medical	reimbursement_landscape	CMS + commercial payer + ICER strategy
Pharma/Medical	regulatory_milestone_playbook	Replaces marketing with milestone-driven growth
Federal Capture	sam_gov_pipeline	SAM.gov registration + opportunity tracking
Federal Capture	gsa_schedule_strategy	MAS + GWACs + alternative vehicles
Federal Capture	set_aside_eligibility	Federal programs + sole-source thresholds
Federal Capture	dfars_defense_compliance	CMMC + NIST 800-171 + Section 889
Federal Capture	federal_capture_playbook	Modifier playbook overlaid on any base sector

Engineering note — the modifier pattern

Federal Contractor Capture is unique: it's a MODIFIER, not a replacement. A construction.commercial_gc firm with 30% federal revenue gets BOTH templates applied — standard commercial GC playbook for its private work + federal capture overlay for its federal work. The Master Router triggers the modifier when the federal_revenue_pct flag is $\geq 15\%$. The other three templates (RIA, Civil/Heavy, Pharma) replace the base profile entirely.

SPECIALTY TEMPLATE — RIA / WEALTH MANAGEMENT — COMPLIANCE-RESTRICTED TEMPLATE

Why a specialty template

Registered Investment Advisers operate under SEC Investment Advisers Act of 1940 and state equivalents. Marketing communications fall under Rule 206(4)-1 (Marketing Rule, effective Nov 2022) which permits testimonials + endorsements but with strict disclosure requirements. Performance advertising rules forbid hypothetical performance presentation without specific safeguards. Most v4 marketing playbook entries (testimonials, social proof, case studies with returns) trigger compliance review. This template replaces the marketing playbook with a compliant advisory growth playbook.

Template ID	specialty.ria_wealth_compliance
Replaces / extends	finance.ria_wealth_management
Business model	B2C_relationship + B2B_fee_based
NAICS examples	523930, 523991, 523999
NAICS-6 routing	['523930', '523991', '523999', '524292']

Compliance filter (mandatory pre-screen for every recommendation)

Every advisory growth action below has been pre-screened against the SEC Marketing Rule, Investment Advisers Act, and FINRA Rule 2210 (where applicable). The compliance flag indicates which compliance review level the recommendation requires before publication.

Compliance flag levels

Flag	Meaning
GREEN	No compliance review needed — operational improvement, no marketing implication
YELLOW	CCO review required — marketing language must be reviewed for compliance
RED	Outside counsel review required — testimonial, performance claim, or endorsement
BLOCKED	Cannot be done compliantly — provided as anti-pattern only

Forbidden recommendation categories

- Hypothetical performance projections
- Cherry-picked performance windows
- Testimonials without required disclosures (advertised + paid status, conflicts of interest)
- Guarantee language ("guaranteed returns", "risk-free", "outperform")
- Past specific recommendations presentation without disclosing all comparable recommendations
- Net-of-fees vs gross-of-fees inconsistency in performance presentation
- Unsubstantiated claims about expertise ("expert in", "leading", "premier")

- Use of "registered" or "approved" implying regulatory endorsement

Data sources

Source	Returns	Cost	Fallback
SEC Investment Adviser Public Disclosure (IAPD)	Form ADV Parts 1+2 — competitor intelligence	Free public	Direct adviserinfo.sec.gov
SEC Form ADV bulk data files	Mass Form ADV download for benchmarking	Free public	Direct
FINRA BrokerCheck	Rep-level disclosure history	Free public	Direct
NASAA + state IA division	State-registered IA data (under-\$100M AUM firms)	Free public	Direct
Cerulli Associates + Schwab Independent Advisor Outlook	Industry benchmarks (AUM growth, fees, technology)	Subscription	Public summary
Charles Schwab RIA Compensation Report + Fidelity RIA Benchmarking	Compensation + practice metrics	Subscription	Public summary
IAA + Investment Adviser Association	Industry association data + benchmarks	Member-only	Public summary
FA Magazine + InvestmentNews + RIABiz	Industry trends + named-firm intelligence	Subscription	Public
Custodians (Schwab Advisor, Fidelity Institutional, Pershing Advisor Solutions, BNY Mellon Pershing)	Tech platform integration data	Direct	Industry
Named TAMPs (Envestnet, Orion Portfolio Solutions, AssetMark, SEI)	Outsourced model + tech intelligence	Direct	Industry

Competitor radius

Mass affluent (\$100k-1M investable assets): 25-50 mi local. HNW (\$1M-10M): regional 200-500 mi. UHNW (\$10M+): national + global. Specialty (executives, divorcees, business owners, physicians): national within specialty.

Review categories tracked

- Communication frequency + responsiveness
- Plan quality + holistic approach
- Fee transparency + value
- Performance vs benchmark (with required disclaimers)
- Investment philosophy clarity

- Tax-efficiency execution
- Estate planning integration
- Crisis communication + market downturn behavior

Advisory Growth Playbook

20 categories — each with research basis + named example. Compliance flags shown where applicable.

Flag	Category	Research basis	Named example
YELLOW	Form ADV Part 2A + 2B optimization	SEC Form ADV brochure + brochure supplement requirements + plain English rule	Annual Form ADV update + plain English rewrite + named CCO review — drives prospect-conversion lift via clear differentiation while staying compliant. Fee schedule + conflict disclosures stay current.
YELLOW	CRS (Customer Relationship Summary) optimization	Form CRS requirements (Reg BI for dual-registered, ADV CRS for IAs)	Form CRS rewrite + CCO review + plain English principles + comparison-friendly format — captures higher prospect conversion vs boilerplate competitor CRS.
GREEN	Niche specialization (executives, physicians, divorcees, business owners)	Cerulli + named specialty firms (e.g. Mercer Advisors physician focus, named family offices)	Niche specialty (e.g. tech executives at named-Fortune-500-employer concentrated equity / RSU expertise) — drives 22-38% higher LTV per client + premium fee structure + organic referral compounding.
GREEN	Designations + credentials (CFP, CFA, ChFC, CPWA, AEP, CDFA)	Named designation programs + CFP Board + CFA Institute + IWI + AICPA	CFP + CPWA + CDFA designations on lead adviser + 2 staff — captures premium positioning + fee floor lift + retention moat. Designation usage on materials must follow CFP Board requirements.
GREEN	Custodian + tech platform consolidation	Schwab Advisor + Fidelity Institutional + Pershing Advisor + named tech (Orion, Black Diamond, Envestnet, eMoney, RightCapital, MoneyGuidePro, Tamarac)	Schwab Advisor + Orion + eMoney integration — drops operational time per client 28% + frees adviser hours for relationship + scales practice from \$80M to \$200M AUM without proportional staff add.
YELLOW	Fee structure optimization	Cerulli + Schwab fee benchmarks + named pricing (AUM, retainer, hourly, project)	Move from pure-AUM to tiered AUM + planning retainer hybrid — drops sub-\$500k client unprofitability + captures HNW segment willing to pay flat planning fee + AUM. Fee schedule update requires ADV amendment.

Flag	Category	Research basis	Named example
YELLOW	COI (Center of Influence) referral pipeline	Named COI relationships (CPAs, attorneys, business brokers, P&C; insurance)	12 named CPA + estate-attorney + business-broker COI nurture — drives 38% of new asset growth at \$0 ad spend. Compensation arrangements (if any) require disclosure on Form ADV.
RED	Solicitor / Promoter agreement (under SEC Marketing Rule)	SEC Marketing Rule (effective Nov 2022) — promoter compensation disclosure requirements	Engage named outside CPA firm as promoter under written agreement + required disclosures + 1099 compensation tracking — drives qualified referrals. Outside counsel review of agreement language required.
RED	Client testimonial + endorsement program (post-Marketing-Rule)	SEC Marketing Rule testimonial framework — requires status disclosure (advertised), conflicts disclosure, qualifying statements	Compliant testimonial program with required disclosures: "This testimonial is given by an advisory client (advertised). [Client] received no compensation. [Adviser] has implemented compliance procedures." Outside counsel must review every testimonial. Most firms find the operational burden too high to scale.
RED	Performance advertising (compliant presentation)	SEC Marketing Rule performance advertising — net-of-fees, comparable benchmark, required time periods	GIPS-compliant composite performance presentation + 1-year + 5-year + 10-year + since-inception + benchmark comparison + net-of-fees + standardized methodology. Most independent RIAs find this operationally impossible without dedicated GIPS verification — typical workaround: avoid performance advertising entirely.
BLOCKED	Hypothetical performance + projections	SEC Marketing Rule prohibits hypothetical performance presentation to retail audiences without unique safeguards	DO NOT show hypothetical "if you invested \$X you would have \$Y" projections to mass-market audiences. This is a Marketing Rule violation regardless of footnote disclosure. Retirement Monte Carlo projections in client-specific plans are different (1:1 advice, not marketing) but cannot be used as marketing material.
YELLOW	Content + thought leadership (educational, not promotional)	Educational content vs promotional content distinction + named platforms	Weekly educational LinkedIn + monthly newsletter on tax planning + retirement income — drives organic credibility. Content must be purely educational; specific recommendations must include "this is general information not personalized advice" disclaimers.

Flag	Category	Research basis	Named example
YELLOW	Digital marketing + paid acquisition	Named platforms (LinkedIn Ads, Meta, Google Ads) + advisor-specific compliance	Targeted LinkedIn Ads to named-employer executives within geographic radius + landing page reviewed by CCO + lead-magnet PDF reviewed by CCO. Avoid generic "best returns" / "top adviser" language — those phrases trigger compliance issues.
YELLOW	Email marketing + drip campaigns	CAN-SPAM + Marketing Rule applicable to email + unsubscribe requirements	Educational drip campaign + monthly market commentary (without specific recommendations) + quarterly newsletter — must have unsubscribe + commercial nature disclosure. Subject lines cannot mislead.
YELLOW	Webinars + educational events	Webinar content treated as advertisement under Marketing Rule	Quarterly educational webinar + recorded version on website — content must be reviewed pre-recording. Recordings must be retained per SEC books-and-records (5 yrs).
YELLOW	Books + media + podcast appearances	Author-status testimonials + podcast appearance compliance	Authored book + podcast guest spots (named industry podcasts: Michael Kitces, Wealthtech on Deck, Barron's Advisor) — drives massive credibility. Book content + podcast commentary must avoid hypothetical performance + specific recommendations to listeners.
GREEN	Acquisition / tuck-in M&A;	Named M&A; + valuation (RIA market currently 8-14× EBITDA + 2-3.5% AUM)	Acquire retiring solo RIA at 10× EBITDA + 5-yr earn-out — captures \$40-200M in AUM at known retention + cross-sell. RIA M&A; is currently the highest-multiple acquisition activity in financial services.
GREEN	Outsourced CCO + compliance technology	Named outsourced CCO firms (RIA in a Box [Compliance Solutions Strategies], MarketCounsel, ACA Group, NRS) + compliance tech (Smarsh, Hearsay, Erado, Global Relay, Mimecast)	Outsourced CCO + Smarsh archiving + named compliance tech — drops compliance cost vs in-house CCO + scales for SEC examination prep. Required for any firm with > \$25M AUM that doesn't have full-time compliance staff.

Flag	Category	Research basis	Named example
GREEN	Cyber + business continuity (Reg S-P + state AI laws)	Reg S-P + named cyber tech (Drata, Vanta, Tessian, ProofPoint) + state SHIELD / CCPA / NY DFS	Drata + named MSP + cyber insurance — drives SEC exam readiness + state law compliance + client trust positioning.
GREEN	TAMP + outsourced investment management	Named TAMPs (Envestnet, Orion Portfolio Solutions, AssetMark, SEI, Buckingham, Brinker)	Envestnet or Orion Portfolio Solutions TAMP — outsources investment management + frees adviser for relationship work + captures higher AUM growth via scale. Trade-off: AUM-fee compression.

KPIs

- Total AUM + AUM growth rate (YoY, target > 12% mature firm, > 25% growth firm)
- Net new asset growth (excluding market growth)
- Asset retention rate (target > 96%)
- Client retention rate (target > 94%)
- Average client AUM (size up over time)
- Revenue per adviser (target > \$1.4M for HNW, > \$800k for mass affluent)
- Operating margin (target > 30% mature firm)
- Adviser productivity (clients per adviser, AUM per adviser)
- NIGO rate (Not In Good Order — process quality)
- NPS / client satisfaction (collected per Marketing Rule compliance)

Cycle / calendar model

Window	Trigger	Action
Annual: Form ADV update	Required by 90 days after fiscal year-end	ADV amendment + brochure update + CCO review
Annual: GIPS verification (if applicable)	GIPS standards require annual third-party verification	External verifier engagement
Quarterly: client review meetings	Performance presentation + plan refresh	Compliant performance materials + Marketing Rule application
Quarterly: net-new-asset target check-in	Internal practice metric	Adviser productivity review
Annual: SEC examination readiness	SEC IA exam every 3-7 yrs typical	Books + records review + CCO mock exam
Annual: cyber risk assessment	Reg S-P requirement + state laws	Cyber insurance renewal + named tech partner audit

Window	Trigger	Action
Quarterly: COI relationship cadence	CPA + attorney + business broker outreach	Lunch + content sharing + introduction tracking
Annual: succession + valuation review	Continuity + buy-sell + retirement	Named M&A; advisor benchmark check + valuation update

Issue library

- SEC Marketing Rule compliance burden — universal — frame as outsourced CCO + named tech
- Form ADV + brochure currency — universal — frame as annual update + plain English rewrite
- Custody rule (Rule 206(4)-2) compliance — universal — frame as named custodian + audit
- Reg S-P privacy + cybersecurity — universal — frame as named MSP + cyber insurance
- Fiduciary duty + best interest standard — universal — frame as documentation discipline + IPS
- Fee compression + AUM-fee pressure — universal — frame as service-fee unbundling + retainer + tax planning
- Adviser succession + buy-sell — universal — frame as named M&A; advisor + valuation
- TAMP vs in-house investment management decision — universal — frame as adviser-time-cost analysis
- State registration requirements (< \$100M AUM) — universal — frame as state IA division + named outsourcing
- Disciplinary disclosure + reputation — universal — frame as transparent ADV + remediation if applicable

Report sections

Section	Contents
Practice snapshot	AUM, advisers, custodian + tech stack, named comp set, ADV summary
Compliance posture audit	ADV currency + Marketing Rule readiness + cyber + custody + Reg BI
Top 5 compliant growth moves	Drawn from advisory_growth_playbook with compliance flags
Niche + designation roadmap	Specialty + named designations + premium positioning
Custodian + tech + TAMP plan	Named platforms + investment priorities
COI + referral pipeline	Named CPA + attorney + business-broker prospects
M&A; + succession positioning	Named M&A; advisors + valuation roadmap
Anti-pattern guide	BLOCKED items: what NOT to do that competitors might suggest

Specificity rules

- Every named regulator must be real (SEC IAPD, FINRA BrokerCheck, NASAA, state IA divisions)
- Every named custodian must be real (Schwab Advisor Services, Fidelity Institutional, Pershing Advisor Solutions, BNY Mellon Pershing, Goldman Sachs Custody, Altruist)
- Every named TAMP must be real (Envestnet, Orion Portfolio Solutions, AssetMark, SEI, Buckingham Strategic Wealth, Brinker Capital)

- Every named tech platform must be real (Orion, Black Diamond by SS&C;, Envestnet Tamarac, eMoney, RightCapital, MoneyGuidePro, Holistiplan, Income Lab, Smarsh, Hearsay, Erado, Global Relay, Mimecast)
- Every named designation must be real (CFP, CFA, ChFC, CFS, AIF, AEP, CPWA, CIMA, CDFA, CRC, RICP)
- Every named outsourced CCO firm must be real (RIA in a Box / Compliance Solutions Strategies, MarketCounsel, ACA Group, NRS, Foreside)
- No recommendation may rely on testimonials, performance projections, or guarantees without explicit BLOCKED flag
- Every recommendation must be compatible with SEC Investment Advisers Act + Marketing Rule (Rule 206(4)-1) at the time of writing

SPECIALTY TEMPLATE — CIVIL / HEAVY CONSTRUCTION — BID INTELLIGENCE TEMPLATE

Why a specialty template

Civil + heavy contractors (highway, utility, water/sewer, dam, bridge, port, federal infrastructure) win work through public bid lettings, not Yelp reviews. Standard v4 marketing playbook (review velocity, social media, lead-gen) doesn't apply. Their growth depends on: (1) bid pipeline visibility, (2) bonding capacity to bid on bigger projects, (3) DBE/MBE/SBE/HUBZone certifications that unlock set-aside categories, (4) EMR < 1.0 to qualify for safety-required work, (5) past performance on federal work (CPARS), (6) joint venture relationships to access work above own bonding capacity. This template replaces marketing playbook with bid pursuit + capacity-building actions.

Template ID	specialty.civil_heavy_bid_intelligence
Replaces / extends	construction.civil_heavy
Business model	B2G_bid + B2B_relationship hybrid
NAICS examples	237110, 237120, 237130, 237210, 237310, 237990
NAICS-6 routing	['237110', '237120', '237130', '237210', '237310', '237990']

Bid pipeline data sources

Source	Returns	Cost	Fallback
SAM.gov (System for Award Management)	Federal opportunities + contracts > \$25k	Free public	Direct sam.gov
FedBizOpps successor: SAM.gov Contract Opportunities	Federal solicitation feed	Free public	Direct
Each state DOT bid letting calendar (e.g. PennDOT, Caltrans, TxDOT, FDOT, GDOT, NYSDOT, IDOT, ODOT, MDT, WisDOT)	State highway + bridge + transit lettings	Free public	Direct state-by-state
DBIA (Design-Build Institute of America) opportunity tracker	Design-build pursuit pipeline	Member-only	Public summary
ConstructConnect / Dodge Construction Network	Pre-bid + plan room access (PAID)	Subscription \$\$\$	BidNet alternative
BidNet Direct + DemandStar	State + local bid aggregators	Subscription	Direct

Source	Returns	Cost	Fallback
AGC (Associated General Contractors) of America + state chapters	Industry benchmarks + local bid sharing	Member-only	Public
ABC (Associated Builders + Contractors) of America	Open-shop benchmarks + advocacy	Member-only	Public
ARTBA (American Road & Transportation Builders Association)	Highway + transportation policy + opportunities	Member-only	Public summary
Federal DOT FHWA + FRA + FAA + USACE + EPA STAG opportunity feeds	Specific federal infrastructure programs	Free public	Direct
IIJA (Infrastructure Investment + Jobs Act) tracker — Build.gov	IIJA-funded project pipeline 2022-2026	Free public	Direct
BABA (Build America Buy America) compliance database	Domestic content requirement tracking	Free public	Direct

Set-aside certification stack

Civil/heavy contractors with the right certifications can bid on set-aside work that competitors cannot. These certifications are the single biggest revenue lever for under-\$50M-revenue firms.

Available certifications

Cert	What it unlocks	Issuer	Renewal
DBE (Disadvantaged Business Enterprise)	Federal DOT-funded work — typically 10-15% of project value reserved for DBEs	State UCP (Unified Certification Program) administered	Annual + 3-yr re-certification
MBE (Minority Business Enterprise)	NMSDC + state/city MBE programs — private sector + government	NMSDC certified — usable across most major corporations	Annual
WBE (Woman-Owned Business Enterprise)	WBENC certification — federal + corporate	WBENC + state programs	Annual
SBE (Small Business Enterprise)	Federal SBA + state programs — size-based not minority-based	SBA Dynamic Small Business Search	Annual
HUBZone (Historically Underutilized Business Zone)	Federal SBA — 35% set-aside on federal contracts	SBA HUBZone certification — geographic + employment requirements	Recertification with map updates

Cert	What it unlocks	Issuer	Renewal
SDVOSB / VOSB (Service-Disabled Veteran-Owned)	Federal — 3% federal contract goal + VA Vets First Verification	VA Office of Small + Disadvantaged Business Utilization	Annual
8(a) Business Development	SBA 9-year program — sole source contracts up to \$4.5M (services) / \$7M (mfg)	SBA 8(a) certification	9-year program lifetime
EDWOSB (Economically Disadvantaged Woman-Owned)	Federal — set-aside in named NAICS	SBA EDWOSB certification	Annual

Stacking strategy

Many certifications stack. A woman-veteran-owned-small-business in a HUBZone can hold WBENC + WOSB + SDVOSB + HUBZone + SBE simultaneously, unlocking 5 different set-aside categories.

Bonding capacity intelligence

Bonding capacity caps how big a project a firm can bid on. Building bonding capacity is one of the slowest + most important growth levers for civil/heavy contractors. Every contractor should know their single project capacity (largest bond surety will write) and aggregate (total simultaneous bond load surety will support).

Named contract sureties

Surety	Profile
Travelers Bond + Specialty Insurance	Largest US contract surety — \$1.5B+ written annually
Liberty Mutual Surety	Top 5 US — strong contractor relationship focus
Zurich North America Surety	Top 5 — strong on infrastructure
Chubb Surety	Top 10 — large project focus
CNA Surety	Top 10 — small + mid-market focus
Westfield Specialty Insurance	Mid-market specialist
Old Republic Surety	Small + mid contractor specialist
HCC Surety Group (Tokio Marine)	Mid-market specialist
IFG Companies	Small-business surety specialist
Argo Surety	Mid + large — IJJA project focus
Markel Surety	Specialty + niche project surety

Capacity expansion levers

- Improve audited financial statements (CPA-reviewed → CPA-audited)
- Stronger working capital position (target > 5% of project value)
- Cleaner balance sheet (lower debt-to-equity)
- Demonstrated project management on prior work
- Personal guarantees from owners (typical for under-\$10M-revenue firms)
- Surety relationship tenure (a 10-yr surety relationship typically = 50% higher capacity vs new)
- Joint venture with larger contractor (uses partner's capacity for the project)

EMR (Experience Modification Rate) intelligence

EMR is workers comp insurance modifier — < 1.0 means below-average claims experience. Many federal + state + private clients require EMR < 0.85 or < 1.0 to bid. EMR is calculated by NCCI (National Council on Compensation Insurance) based on 3-year claim history.

EMR benchmarks

EMR range	Implication
EMR > 1.20	High risk — likely disqualified from many bids
EMR 1.00-1.20	Industry average — competitive but not premium
EMR 0.85-1.00	Below average — qualifies for most safety-required work
EMR 0.70-0.85	Top quartile — premium positioning + insurance savings
EMR < 0.70	Top decile — preferred contractor status

Reduction levers

- Documented safety culture program (named: PPE compliance, daily JSA, weekly toolbox)
- OSHA 30-hour training for supervisors + 10-hour for craftworkers
- Drug-free workplace program with named provider (Quest Diagnostics, LabCorp)
- Return-to-work + light-duty program to minimize claim cost
- Named safety-management software (Procore Quality + Safety, SafetyCulture iAuditor, HCSS HeavyJob safety module, Smartvid.io)
- AGC + ABC safety training programs (CHASM, STEP)
- EMR appeals process (named claims that should not have applied)

CPARS past performance strategy

CPARS (Contractor Performance Assessment Reporting System) is the federal database of past contract performance. Federal contracting officers must consider CPARS scores in award decisions. CPARS scores affect every future federal bid.

CPARS evaluation categories

- Quality of product or service
- Schedule (on-time, milestone performance)
- Cost control
- Management (incl. business relations)

- Utilization of small business (small business participation)
- Regulatory compliance

Rating scale: Exceptional / Very Good / Satisfactory / Marginal / Unsatisfactory

Improvement levers

- Pre-award discussion with CO about expectations
- Quarterly progress reports volunteered (pre-empt CO requests)
- Document delays attributable to government (issue resolution requests, modifications)
- Post-award meet-and-greet with COR (Contracting Officer's Representative)
- Named small business subcontractors (drives Utilization rating)
- Formal dispute prevention via name / position changes when CO request seems unreasonable

Competitor radius

Local: 50-100 mi for utility + small civil. State: state DOT-defined for highway + bridge work. Federal: nationwide for federal infrastructure. IIJA + BABA-funded work: defined by named program (e.g. Bridge Investment Program is multi-state pursuit, EV Charging Infrastructure is statewide).

Bid Pursuit Playbook

15 categories — each with research basis + named example. Compliance flags shown where applicable.

Category	Research basis	Named example
Set-aside certification stack maximization	Owner profile + named certifying bodies (UCP, NMSDC, WBENC, SBA, VA)	Owner is woman + minority + Vietnam-era veteran in HUBZone — pursue WBENC + NMSDC + SDVOSB + HUBZone simultaneously — unlocks 4 set-aside categories. Investment ~\$4-12k + 6-9 mo for full stack.
Bonding capacity expansion program	Named surety relationships + audited financial readiness + working capital targets	Move from CPA-reviewed to CPA-audited financials + improve working capital ratio to 8% + named surety relationship deepening with Travelers Bond — captures bonding capacity from \$10M single / \$30M aggregate to \$25M / \$80M over 18 mo.
EMR reduction safety program	Named NCCI EMR formula + AGC STEP / ABC CHASM safety programs	AGC STEP Diamond + named safety software + Quest Diagnostics drug-free workplace — drops EMR from 1.10 to 0.85 over 3 yrs, qualifies for safety-required federal + utility work + drops insurance premium 30%+.
CPARS rating improvement	Named federal contracting officers + pre-award + post-award discipline	Pre-award meeting + named-CO relationship + voluntary quarterly progress reports + named small business subcontractor utilization — moves CPARS scores from Satisfactory to Very Good over 3 contracts, changes future bid award math.

Category	Research basis	Named example
IIJA-funded project pursuit pipeline	Build.gov tracker + state DOT IIJA spending plans + BABA compliance	IIJA programs by funding stream (Bridge Investment, Reconnecting Communities, EV charging, lead pipe replacement) + state DOT 4-yr spending plan + BABA domestic content compliance — captures share of \$1.2T infrastructure investment 2022-2026.
Joint venture + teaming agreement strategy	Named regional + national contractors + teaming agreement structure	JV with named larger contractor (e.g. Kiewit, Granite, Skanska, Lane, Fluor, AECOM, Jacobs) on \$50M+ project — uses partner's bonding + experience while smaller firm builds CPARS + capacity. Eventually graduates to prime on similar work.
GSA Schedule + multiple-award task order contracts (MATOC) registration	Named federal contract vehicles (GSA MAS, USACE MATOC, NAVFAC MACC, DOE IDIQ)	GSA MAS Schedule for environmental + construction services + USACE MATOC qualification — captures multi-year IDIQ work + reduces per-bid pursuit cost dramatically.
Pre-bid + plan room intelligence subscription	Named platforms (ConstructConnect, Dodge Construction Network, BidNet, ISqFt, Procore Bid Management)	ConstructConnect Insight subscription — captures 90+ days advance project intelligence + named owner + GC contacts + drives proactive teaming + outpaces reactive-bid competitors.
BABA + Buy America compliance readiness	Build America Buy America Act + named domestic supplier networks	BABA compliance documentation + named domestic steel + iron + manufactured product suppliers — captures BABA-funded work (most IIJA \$) + drives premium positioning vs offshore-sourcing competitors.
Federal small business + 8(a) program participation	SBA 8(a) Business Development + Mentor-Protégé + sole-source thresholds	8(a) certification + sole-source contracts up to \$4.5M (services) / \$7M (mfg) + 9-yr program participation — captures non-competitive federal work + accelerates capability building.
Past project portfolio documentation	Project resume formatting + named completed work + photographs + references	Standardized project resume database + 10 named past projects with outcomes + 5 owner references + project photographs — captures bid evaluation points + drives award decisions.
Estimating + bid management software	Named estimating platforms (HeavyBid by HCSS, B2W Estimate, Bid2Win, Sage Estimating, BluebeamRevu Q+E)	HCSS HeavyBid + integrated takeoff + price-database + named subcontractor invitations — drops bid preparation time 40% + drives bid hit rate from 12% to 18% via more bids submitted.

Category	Research basis	Named example
Equipment + fleet capability documentation	Named equipment manufacturers + asset documentation	Owned equipment list (Cat, John Deere, Komatsu, Volvo CE, Liebherr) + maintenance history + GPS / telematics — captures owner-equipment-required bids + improves bid scoring.
Project labor agreement (PLA) + union readiness	Named building trades unions + PLA project pipeline	Building Trades Council relationship + named PLA-required project tracking — captures PLA-required work (much federal + many state DOT projects post-2022 EO 14063) that open-shop competitors cannot bid.
M&A; + bolt-on acquisition strategy	Named M&A; advisors + civil contractor valuation (4-7× EBITDA typical)	Acquire smaller civil contractor with complementary geography + cert stack + bonding history — adds CPARS portfolio + immediate certs + bonding capacity at 5× EBITDA. Civil consolidation is active in IIJA-driven environment.

KPIs

- Backlog (\$) + months of work
- Bid hit rate (bids won / bids submitted, target 15-25%)
- Avg project margin variance (estimated vs actual gross margin)
- EMR (target < 0.85)
- OSHA recordable rate (TRIR target < 1.5)
- Bonding capacity — single project + aggregate
- Set-aside cert count (active + pending)
- CPARS rating average (target > Very Good)
- IIJA + BABA-funded work as % of revenue
- Repeat client revenue % (target > 55%)

Cycle / calendar model

Window	Trigger	Action
Federal FY-end (Sep 30)	Federal spending push + use-or-lose obligations	Bid frequency + pursuit max
State DOT bid letting calendar	Monthly + quarterly lettings (state-specific)	Pre-bid attendance + question submission + bid prep
Capital budget season (Aug-Nov)	Annual capital budget approval	Pre-construction engagement + early bidder relationships
Construction season (Apr-Oct in north)	Project execution peak	Schedule + production + safety + CPARS performance

Window	Trigger	Action
Annual: surety + insurance renewal	Bond capacity review + EMR update	Renewal discipline + capacity expansion
Annual: DBE/MBE/SBE re-certification	Cert renewal cycles	UCP renewal + NMSDC + SBA
Annual: CPARS narrative review	Federal CPARS database refresh	Disputed rating remediation + new project documentation
IIJA program rollout (2022-2026)	Program-by-program funding deployment	Tracker subscription + named-program pursuit

Issue library

- Bonding capacity caps bid size — universal — frame as named surety + capacity-building program
- EMR > 1.0 disqualifies from work — universal — frame as AGC STEP + named safety software
- Set-aside cert pursuit complexity — universal — frame as named UCP + advisor support
- CPARS scores affecting future awards — universal — frame as pre-bid + post-award discipline
- BABA compliance + domestic supply chain — universal — frame as named domestic supplier + documentation
- Estimating accuracy + margin variance — universal — frame as HeavyBid + production-rate database
- Subcontractor + supplier reliability — universal — frame as approved-vendor list + payment performance
- IIJA opportunity capture — universal — frame as Build.gov tracker + state plan engagement
- Joint venture relationship dynamics — universal — frame as named teaming + revenue-share clarity
- Field labor + craftworker shortage — universal — frame as apprenticeship + named technical college

Report sections

Section	Contents
Firm snapshot	Equipment + crew + bonding + named comp set
Set-aside cert stack audit + roadmap	Active + pursuable certifications
Bonding capacity expansion plan	Named surety + financial + working capital
Top 5 bid pursuit moves	Drawn from bid_pursuit_playbook
EMR + safety + insurance roadmap	Named AGC STEP + safety tech + insurance
CPARS + past performance plan	Federal contract performance trajectory
IIJA + BABA opportunity pipeline	Named programs + domestic-content prep

Section	Contents
Joint venture + M&A; positioning	Named teaming + acquisition opportunities

Specificity rules

- Every named federal program must be real (IIJA, BABA, BIL, EPA STAG, FHWA, FRA, USACE, NAVFAC, GSA MAS, USACE MATOC)
- Every named cert program must be real (DBE/UCP, NMSDC MBE, WBENC, SBA 8(a), HUBZone, SDVOSB/VOSB)
- Every named surety must be real (Travelers Bond, Liberty Mutual Surety, Zurich, Chubb, CNA, Westfield, Old Republic, HCC, IFG, Argo, Markel)
- Every named software platform must be real (HCSS HeavyBid + HeavyJob, B2W Estimate, Bid2Win, Sage Estimating, Procore Bid Management, ConstructConnect, Dodge, BidNet, BluebeamRevu)
- Every named contractor (for JV examples) must be a verifiable currently-active contractor
- Every named industry program must be real (AGC STEP, ABC CHASM, NUCA, ARTBA programs)
- No marketing recommendations — bid pursuit + capacity-building only

SPECIALTY TEMPLATE — PHARMA / MEDICAL DEVICE — REGULATED LIFECYCLE TEMPLATE

Why a specialty template

Pharma + medical device companies grow by passing FDA milestones, not by closing more sales. A drug company with great Phase 3 data but no marketing team is worth billions. A drug company with great marketing but failed Phase 3 data is worth zero. Standard v4 playbook (review velocity, lead-gen, content marketing) is irrelevant. This template instead tracks: FDA submission lifecycle (IND/NDA/BLA for drugs; 510(k)/PMA/De Novo for devices), CRO + CMO partnerships, quality system compliance (cGMP, ISO 13485, 21 CFR 820), reimbursement landscape (CMS, payers, ICER), patent + IP strategy, manufacturing scale-up. KPIs are pipeline-driven, not revenue-driven.

Template ID	specialty.pharma_medical_regulated
Replaces / extends	manufacturing.advanced_regulated (subset: pharma + medical device only)
Business model	B2B_regulated_lifecycle (multi-year, milestone-driven)
NAICS examples	325412, 325413, 325414, 339112, 339113
NAICS-6 routing	['325411', '325412', '325413', '325414', '339112', '339113', '339114', '339115',...

FDA submission lifecycle

FDA submission pathway determines timeline, cost, risk, and ultimate market access. Different products have radically different lifecycles.

Drug pathway

Stage	Activity	Timeline	Cost
Pre-IND	Discovery + lead optimization	Years -5 to -1	\$2-50M cost
IND (Investigational New Drug)	Pre-clinical → first-in-human	Year 0-1	\$15-50M to file
Phase 1 — Safety	20-100 healthy volunteers	Year 1-2	\$10-50M
Phase 2 — Efficacy	100-500 patients	Year 2-4	\$30-200M
Phase 3 — Pivotal	1000-5000 patients, multi-site	Year 4-7	\$100M-2B
NDA / BLA submission	New Drug Application or Biologics License Application	Year 6-7	\$2-5M filing
FDA Review	Standard 10mo / Priority 6mo / Breakthrough designation	Year 7-8	PDUFA fee \$4M+

Stage	Activity	Timeline	Cost
Approval + launch	Manufacturing scale-up + commercial launch	Year 8+	Variable
Phase 4 / PMR	Post-marketing requirements + REMS	Ongoing	Variable

Device pathway — 510(k)

Stage	Activity	Timeline	Cost
Predicate device research	Identify substantially-equivalent predicate	Year 0	Internal
Pre-Submission (Q-Sub)	FDA pre-submission meeting	Year 0-1	\$2-15k
510(k) submission	Substantial equivalence pathway	Year 1	\$5-20k FDA fee + \$50-500k prep
FDA Review	90-day target (often 6-12mo actual)	Year 1-2	FDA fee
Clearance + market	Commercial launch	Year 1.5+	Variable

Device pathway — PMA

Stage	Activity	Timeline	Cost
Pre-IDE	Investigational Device Exemption pre-submission	Year 0-1	\$2-15k
IDE filing	Approval to begin clinical trials	Year 1	\$10-50k FDA + \$200k-2M prep
Pivotal clinical trial	Patient + safety + efficacy data	Year 2-5	\$10-100M
PMA submission	Pre-Market Approval application	Year 5-6	\$300-500k FDA + \$1-5M prep
FDA Review + advisory panel	180-day target + advisory committee	Year 6-7	PDUFA fee
Approval + launch	Commercial launch	Year 7-8	Variable

Device pathway — De Novo

Stage	Activity	Timeline	Cost
Pre-Sub for De Novo classification	Novel device with no predicate	Year 0-1	Internal
De Novo Classification Request	Risk-based reclassification pathway	Year 1-2	\$10-25k FDA + \$100-500k prep

Stage	Activity	Timeline	Cost
FDA Review	120-day target	Year 2	PDUFA
Granted + becomes predicate	Future similar devices use 510(k)	Year 2-3	Variable

CRO / CDMO partner landscape

Most pharma + device firms outsource clinical research (CRO) and contract manufacturing (CMO/CDMO). Choosing the right partner determines timeline + cost + regulatory risk.

Drug CROs

Partner	Profile
IQVIA	Largest CRO globally — full service
Labcorp Drug Development (formerly Covance)	Full service + central labs
ICON plc	Top 5 CRO — full service
Parexel	Top 5 — global trials specialist
Charles River Laboratories	Pre-clinical + early development specialist
PPD (Thermo Fisher)	Full service
Syneos Health	Mid + commercial-stage
Medpace	Mid-size — therapeutic-area specialist
WuXi AppTec	Global — China-based
Veeva Systems	Clinical + reg tech (not full CRO)

Device-focused CROs

Partner	Profile
NAMSA	Medical device CRO specialist
Eurofins Medical Device Testing	Device testing + analytical
Intertek Medical Device	Device testing + cert
Boston Scientific iLab (in-house)	Various — by major OEM
SGS Health Sciences	Multi-modal

Drug CDMOs

Partner	Profile
Catalent	Largest CDMO — drug + biologics
Lonza	Top 3 — biologics + cell + gene therapy
Thermo Fisher Scientific (Patheon)	Top 3 — full service
Recipharm	Mid-large — sterile + solid dose
CordenPharma	Mid — API + complex dose
Vetter Pharma	Sterile injectable specialist
Samsung Biologics	Biologics specialist
Boehringer Ingelheim BioXcellence	Biologics CDMO arm
Fujifilm Diosynth Biotechnologies	Biologics specialist

Device CMOs

Partner	Profile
Jabil Healthcare	Top device CMO
Flex Health Solutions	Top device CMO
Sanmina	Mid — full service
Plexus	Mid + complex devices
Integer Holdings	Implantable device specialist
Phillips-Medisize (Molex)	Device + drug delivery

Quality system compliance

FDA + EU + global regulators require documented quality systems. The standards are well-defined. Compliance is non-optional.

Frameworks

Framework	Scope
cGMP (Current Good Manufacturing Practices) — 21 CFR 210/211	Drug manufacturing baseline + FDA inspections
21 CFR 820 (Quality System Regulation)	US medical device baseline + FDA inspections
ISO 13485:2016	Global medical device standard + EU MDR + Health Canada + Japan PMDA

Framework	Scope
ISO 14971	Medical device risk management
IEC 60601 series	Electromedical equipment safety
IEC 62304	Medical device software lifecycle
ISO 14644	Cleanroom classification (5/7/8)
EU MDR 2017/745	EU medical device regulation (replaces MDD)
MDSAP (Medical Device Single Audit Program)	US/Canada/Australia/Brazil/Japan single audit

Named QMS platforms

Platform	Profile
MasterControl	Most adopted QMS for pharma + device
Veeva Vault QMS	Cloud-native QMS — pharma standard
Sparta TrackWise	Enterprise QMS — large pharma
ETQ Reliance	Mid-market QMS
Greenlight Guru	Device-focused QMS for startups + mid
MasterControl Manufacturing Excellence	Electronic batch records
Werum PAS-X	MES for pharma manufacturing
Apprentice.io	AI-augmented digital quality

Reimbursement + payer landscape

Even an FDA-approved drug or device fails commercially without reimbursement. Reimbursement strategy must be planned years before approval.

US payers

Payer	Profile
CMS (Centers for Medicare & Medicaid Services)	Largest single payer — Medicare Part B/D + Medicaid
MACs (Medicare Administrative Contractors)	Regional Medicare coverage decisions
Commercial payers (UnitedHealthcare, Anthem/Elevance, Aetna/CVS, Cigna, Humana, BCBS Federation)	Commercial coverage

Payer	Profile
ICER (Institute for Clinical and Economic Review)	Independent value assessment — influences payer decisions
Premier + Vizient + HealthTrust	GPO contracting (devices)

Reimbursement codes

Code system	Use
CPT codes	Procedure billing — AMA-administered
HCPCS Level II	Devices + supplies + services
ICD-10-CM/PCS	Diagnoses + procedures
NDC (National Drug Codes)	Drug products
DRG (Diagnosis-Related Groups)	Inpatient bundled payment

Levers

- Phase 3 trial design with patient-reported outcomes (drives payer value story)
- Real-world evidence (RWE) registry to support post-launch coverage
- Health economics + outcomes research (HEOR) team with named publications
- Payer-advisory boards pre-launch
- Patient assistance + co-pay program design
- CMS NCD (National Coverage Determination) pursuit for breakthrough devices

IP + patent strategy

Patent runway determines commercial value. Composition-of-matter patents protect drugs; method-of-use + formulation patents extend exclusivity.

Lever categories

- Composition of matter (drug substance)
- Method of use / treatment
- Formulation + delivery
- Manufacturing process
- Combination products
- Patent term extension (Hatch-Waxman, BPCIA for biologics)
- Regulatory exclusivity (5-yr NCE, 7-yr orphan, 12-yr biologics, 6mo pediatric)

Named IP firms (pharma)

- Finnegan Henderson (top pharma IP firm)
- Fish & Richardson
- Wilson Sonsini Goodrich & Rosati

- Goodwin Procter (life sciences focus)
- Cooley LLP (life sciences)
- Wilmer Hale
- Sterne Kessler Goldstein & Fox
- Foley Hoag

Data sources

Source	Returns	Cost	Fallback
FDA Drugs@FDA + 510(k) database + PMA database + DMF database	Submission histories + competitor intelligence	Free public	Direct fda.gov
ClinicalTrials.gov	Clinical trial registry + competitor tracking	Free public	Direct
EU Clinical Trials Register + ICTRP	Global trial registry	Free public	Direct
ICER reports	Health economic + value assessments	Free public	Direct icer.org
CMS NCD + LCD database	Coverage decisions	Free public	Direct
SEC filings (8-K, 10-K, 10-Q)	Public-company milestone disclosures	Free public	EDGAR
Cortellis (Clarivate) + Citeline (Informa)	Commercial pharma intelligence	Subscription \$\$\$	Direct
Definitive Healthcare	Provider + facility + reimbursement intel	Subscription	Direct
IQVIA MIDAS + Symphony Health	Prescription + dispensing data	Subscription \$\$\$	Direct
MedTech Insight + Endpoints News + STAT	Industry intelligence	Subscription	Public summary

Competitor radius

Global. Pharma + device markets are global with regulatory + reimbursement variation by jurisdiction.

Regulatory Milestone Playbook

15 categories — each with research basis + named example. Compliance flags shown where applicable.

Category	Research basis	Named example
IND filing readiness + pre-IND meeting	FDA pre-IND meeting strategy + Type B vs C	FDA pre-IND meeting + named CRO (Charles River for tox + IQVIA for clinical) + complete IND-enabling toxicology — drives clean IND filing + minimizes hold/delay risk.
Breakthrough Therapy / Fast Track / Priority Review designation	FDA expedited program criteria + named drug examples	Pursue Breakthrough Therapy designation post-Phase 1 — captures rolling NDA review + ~4-6 mo accelerated timeline + premium investor signaling. Requires substantial improvement vs available therapies on clinically significant endpoint.
Orphan Drug Designation pursuit	FDA orphan criteria (< 200k US patients) + 7-yr exclusivity + tax credit	Orphan designation + 7-yr exclusivity + R&D; tax credit + PDUFA fee waiver — captures premium positioning for rare disease drugs. Filed pre-IND or pre-NDA.
510(k) predicate strategy + Q-Sub	FDA 510(k) database research + predicate selection	Identify multiple potential predicates + Q-Sub to align on substantial equivalence path — drives 6mo timeline reduction + de-risks 510(k) clearance vs predicate-pivot mid-review.
CDMO/CMO selection + qualification	Named CDMOs + capacity audit + tech-transfer planning	Catalent or Lonza CDMO partnership + named QA-led tech transfer + capacity reservation for commercial-scale — de-risks supply at launch + drives manufacturing scalability.
CRO selection + clinical operations build-out	Named CROs + therapeutic-area specialty + site network	IQVIA or Medpace CRO + named site network + medical monitor + statistician — drives Phase 2/3 quality + on-time enrollment + clean database lock.
Patent portfolio strategy + freedom-to-operate	Named IP counsel + composition + method + formulation claims	Composition + method-of-use + formulation patents filed + global PCT filings + named IP firm freedom-to-operate analysis — drives 20-yr commercial exclusivity + acquisition value.
Real World Evidence (RWE) + post-launch registry	FDA RWE framework + named registry vendors (IQVIA, Flatiron Health, Tempus, Aetion)	Patient registry + RWE capability + named vendor partnership — drives label expansion submissions post-launch + payer coverage strengthening + drives FDA Sentinel/EUDRA compliance.
Reimbursement + payer-advisor board (pre-launch)	Named HEOR consultants (Boston Healthcare Associates, Avalere, Trinity Life Sciences) + payer panels	Pre-launch payer advisory board + HEOR dossier + ICER engagement + named CMS conversation — drives launch coverage rate from 40% to 75% + faster patient access.

Category	Research basis	Named example
EU MDR + global regulatory pathway	EU MDR 2017/745 + named EU notified bodies (BSI, TÜV SÜD, Dekra, NSAI, MDC)	EU MDR notified body engagement + UDI registration + technical documentation per Annex II — captures EU market access + drives 20-30% of revenue post-Brexit + post-MDR transition.
Quality management system implementation	Named QMS platforms + ISO 13485 + cGMP	MasterControl or Veeva Vault QMS + ISO 13485 cert + cGMP readiness — drives FDA inspection-ready posture + supports commercial scale + global submissions.
Manufacturing scale-up + tech transfer	Named CDMOs + capacity planning + commercial-batch validation	Validation Master Plan + IQ/OQ/PQ + 3-batch process validation at commercial scale + named CDMO tech transfer — drives launch-readiness + zero supply disruption.
Strategic licensing + business development	Named investment banks (Centerview, Evercore, MTS, Stifel, Cowen) + valuation benchmarks	Out-license non-core program to strategic + retain core program + use proceeds to fund pivotal — captures non-dilutive funding + named-partner validation + drives platform credibility.
M&A; + acquisition by strategic	Pharma + device M&A; multiples (4-12× revenue, 8-25× EBITDA) + named acquirers	Strategic acquisition by big pharma (named: Pfizer, Merck, J&J, Lilly, AstraZeneca, BMS) or device strategic (Medtronic, Stryker, Boston Scientific, Abbott, J&J; Medtech) at validated milestone — captures 5-15× revenue valuation. Most pharma + device value realized via M&A; vs IPO.
IPO + capital strategy	Named bookrunners (Goldman Sachs, JPM, Morgan Stanley, Citi, BofA, Cowen, SVB, Jefferies)	IPO at named milestone (often Phase 2 readout for pharma) + named bookrunners + analyst relationships — captures public-market capital + currency for further development.

KPIs

- Pipeline programs by phase (Phase 1, 2, 3, registration)
- Cash runway (months) at current burn
- IND/NDA/BLA/510(k)/PMA submissions per year
- Regulatory milestone hit rate (target > 80% within projected timeline)
- Patent count + composition-of-matter claims + freedom-to-operate position
- CRO + CDMO partner performance (named scorecards)
- Quality system audit findings (FDA + EU + ISO)
- Clinical trial enrollment rate (% of plan)
- Revenue (post-launch) vs analyst consensus
- Coverage % of US lives + Medicare + commercial mix

Cycle / calendar model

Window	Trigger	Action
Quarterly: pipeline review + milestone update	Internal + investor	Pipeline KPI update + investor 8-K coordination
Annual: FDA inspection readiness	cGMP + 21 CFR 820 audits	Pre-inspection mock + CAPA closure
Multi-year: program lifecycle	5-15 yrs drug, 3-7 yrs device	Long-cycle planning + funding strategy
Annual: ICER + payer engagement cycle	Pre-launch + commercial	HEOR dossier update + payer advisory board
Patent lifecycle: 20-yr clock	Composition + method patents	Patent term extension + lifecycle management
Submission cycles (FDA + EU + global)	510(k) 90d / NDA 10mo / PMA 180d / EU CE 6-12mo	Submission management + regulatory strategy
Annual: CRO + CDMO contract review	Capacity + price + quality	Vendor governance + named-partner discipline

Issue library

- Cash runway vs milestone pacing — universal — frame as strategic licensing + non-dilutive + bridge financing
- CRO performance + database delays — universal — frame as named-CRO scorecards + alternates
- CMC + manufacturing scale-up risk — universal — frame as named CDMO + tech transfer discipline
- Regulatory submission delays — universal — frame as Q-Sub + Type B/C meetings + transparent FDA dialogue
- IP litigation + competitor patents — universal — frame as named IP counsel + freedom-to-operate
- Reimbursement coverage gaps post-launch — universal — frame as HEOR + RWE + payer engagement
- EU MDR transition burden — universal — frame as named notified body + dedicated MDR resources
- Talent retention (RA/QA + clinical ops) — universal — frame as comp + culture + named designations
- M&A; timing + valuation — universal — frame as named investment bank + milestone-driven sale process

Report sections

Section	Contents
Pipeline + asset snapshot	Programs by phase, indications, IP runway, FDA pathway, named comp set
Top 5 regulatory milestone moves	Drawn from regulatory_milestone_playbook

Section	Contents
CRO + CDMO partner roadmap	Named partners + performance + alternates
Quality + cert + audit readiness	Named platforms + audit prep + regulatory dialogue
IP + freedom-to-operate plan	Named IP counsel + patent portfolio + lifecycle mgmt
Reimbursement + commercial-launch readiness	HEOR + RWE + named payer engagement
Capital + strategic + M&A; positioning	Named bankers + acquirer landscape + milestone-driven valuation
Anti-pattern guide	Common mistakes: skipping pre-submission meetings, undertested CMC, late payer engagement

Specificity rules

- Every named regulator must be real (FDA, EMA, MHRA, PMDA, NMPA, Health Canada, TGA, ANVISA, Swissmedic)
- Every named CRO must be real (IQVIA, Labcorp Drug Development, ICON, Parexel, Charles River, PPD/Thermo Fisher, Syneos, Medpace, WuXi AppTec, NAMS, Eurofins, Intertek)
- Every named CDMO must be real (Catalent, Lonza, Thermo Fisher Patheon, Recipharm, CordenPharma, Vetter, Samsung Biologics, Boehringer Ingelheim, Fujifilm Diosynth, Jabil Healthcare, Flex Health, Sanmina, Plexus, Integer Holdings)
- Every named QMS platform must be real (MasterControl, Veeva Vault QMS, Sparta TrackWise, ETQ Reliance, Greenlight Guru, Werum PAS-X, Apprentice.io)
- Every named cert must be real (cGMP 21 CFR 210/211, 21 CFR 820 QSR, ISO 13485, ISO 14971, IEC 60601, IEC 62304, ISO 14644, EU MDR 2017/745, MDSAP)
- Every named acquirer must be a verifiable currently-active pharma or medtech strategic
- No marketing recommendations apply — regulatory milestone + capital + IP only
- Timeline + cost ranges are typical industry ranges; actual program economics vary materially

SPECIALTY TEMPLATE — FEDERAL CONTRACTOR CAPTURE — MODIFIER TEMPLATE

Why a specialty template

Federal contractors operate in a parallel ecosystem from commercial businesses. SAM.gov is their CRM. GSA Schedule is their store. Federal CO + COR are their accounts. CPARS is their review system. A construction firm doing federal work + commercial work needs BOTH the standard construction playbook AND the federal contractor playbook simultaneously. This template is designed to be applied as a modifier — it provides additional capture-pipeline + business-development capability on top of any sector profile. Triggered when $\geq 15\%$ of a firm's revenue comes from federal sources.

Template ID	specialty.federal_contractor_capture
Replaces / extends	NONE — this is a modifier applied alongside a base sector profile
Business model	B2G_capture (federal contracting overlay)
NAICS examples	Any NAICS — modifier applies across sectors
NAICS-6 routing	Triggered by federal-revenue-percentage flag, not NAICS

SAM.gov + federal opportunity pipeline

SAM.gov (System for Award Management) is the unified federal contracting portal. Every contractor must register. Every opportunity is posted there. Pipeline visibility starts here.

Core registration components

Component	Detail
SAM.gov registration + entity profile	Annual renewal + UEI (Unique Entity Identifier) + CAGE Code
NAICS code self-coding	Up to 5 primary NAICS codes — drives opportunity matching
Capability statement (linked from SAM)	Public capability summary for COs to browse
Past performance documentation in SAM	Federal CPARS scores visible to all COs
Representations + Certifications (FAR Part 4)	Annual reps + certs — drives eligibility
Set-aside designations claimed	SBA-certified vs self-certified small business

Opportunity alert + intel

- SAM.gov saved searches by NAICS + keyword + agency
- BetaSAM contract opportunities feed (legacy FedBizOpps successor)

- Agency-specific portals (e.g. NAICS for Navy, Army, Air Force, GSA, DOE, NASA, EPA, DHS, VA, USDA)
- GSA eBuy for GSA Schedule task orders
- NIH NITAAC for IT contracts
- Named third-party aggregators (GovWin IQ by Deltek, Bloomberg Government, govSpend, FedScoop)

GSA Schedule + multiple-award contract strategy

GSA MAS (Multiple Award Schedule) is a long-term contract that lets agencies buy from a contractor without re-competing. Getting on a GSA Schedule typically takes 12-18 months and ~\$50-150k in proposal costs, but unlocks ~\$45 billion in annual federal spend.

MAS categories

Category	Scope
IT Schedule 70 (now subsumed into MAS)	IT products + services
Professional Services (formerly Schedule 738/874)	Mgmt consulting, R&D, engineering
Facilities (formerly Schedule 03FAC)	Janitorial + facility + landscape
Office Management	Office supplies + equipment
Furniture + Furnishings	Office + facility furniture
Industrial Products + Services	Industrial supply + maintenance
Security + Protection	Guard + alarm + protective services
Travel	Travel-related products + services
Transportation + Logistics	Freight + warehousing + courier

Alternative contract vehicles

Vehicle	Use
NASA SEWP (IT)	NASA-administered IT GWAC — \$50B+ ceiling
NIH NITAAC CIO-CS	IT health + commodity
NIH NITAAC CIO-SP3	IT services
DHS EAGLE II	DHS IT services
GSA OASIS	Professional services GWAC
GSA Alliant 2	IT services GWAC
NAVFAC MACC	Navy facilities construction
USACE MATOC	Army Corps construction

Vehicle	Use
CIO-SP4 (NIH)	Successor to CIO-SP3

Federal set-aside eligibility

Federal small-business set-asides reserve specific contracts for qualifying firms. Different programs reserve different categories. Stacking is allowed.

Programs

Program	Eligibility + benefit
Small Business (SB)	Size standards by NAICS (employees or revenue) — federal goal 23%
HUBZone	Geographic + employment requirements — federal goal 3%
8(a) Business Development	Disadvantaged owners + 9-yr program — federal goal 5%
WOSB (Woman-Owned Small Business)	Federal goal 5%
EDWOSB (Economically Disadvantaged)	Subset of WOSB — eligible for sole-source up to thresholds
SDVOSB (Service-Disabled Veteran-Owned)	Federal goal 3%
VOSB (Veteran-Owned)	Most beneficial at VA (Vets First Verification)
Indian Tribes + Alaska Native Corporations + NHO	Special sole-source authority
Mentor-Protégé Program	Pairs small + large firms for capability transfer

Sole-source thresholds (2024)

Program	Threshold
8(a) services	Up to \$4.5M sole source
8(a) manufacturing	Up to \$7M sole source
SDVOSB sole source	Up to \$4M services / \$7M mfg
HUBZone sole source	Up to \$4.5M services / \$7M mfg
WOSB / EDWOSB sole source	Up to \$4.5M services / \$7M mfg (in specific NAICS)

CPARS past performance strategy

CPARS (Contractor Performance Assessment Reporting System) drives every future federal award. CO must consider past performance. Bad ratings haunt for 6 years.

CPARS rating categories

- Quality of product or service
- Schedule
- Cost control
- Management
- Small business utilization
- Regulatory compliance

Improvement levers

- Pre-award meet + greet with CO + COR
- Quarterly status reports volunteered
- Document government-caused delays via formal modifications
- Rebut unfair ratings within 60-day comment period
- Use named small business subcontractors (drives Utilization rating)
- Maintain CPARS narrative — annual updates

Capability statement

A 1-2 page capability statement is the federal contractor's business card. Every CO + COR relationship starts with reviewing this. Should include: company overview, core competencies, differentiators, past performance highlights, codes (NAICS, PSC, CAGE, UEI), set-aside designations, key personnel, contact.

Best practices

- Tailor to specific agency or NAICS (don't use generic for everyone)
- Include CAGE Code + UEI prominently
- List 3-5 most relevant past performance contracts with values
- Use government-friendly language (FAR-aligned)
- PDF format + accessible (Section 508 compliant)
- Short — 1 page front + back, max 2 pages

Teaming + JV strategy

Most large federal contracts go to teams. Small firms bring set-aside eligibility + agility; large firms bring capacity + past performance. Teaming agreements + JVs are critical.

Arrangement types

Type	Detail
Prime + Sub (subcontractor agreement)	Standard arrangement — prime gets paid, subs paid by prime
Joint Venture (JV)	Two firms create new entity to bid as prime — can claim either firm's set-aside if structured properly

Type	Detail
Mentor-Protégé Joint Venture	SBA-approved JV between large mentor + small protégé — JV claims protégé's small-business status without size aggregation
Contractor Team Arrangement (CTA)	GSA Schedule-specific — multiple Schedule-holders combine

Considerations

- Limitation on subcontracting (typically 50% of work must be done by prime + similarly-situated subs)
- NDA + non-circumvention agreements before sharing
- Past performance attribution — JV gets credit for both firms
- Bonding: prime's capacity, with sub indemnification typically

DFARS + defense-specific compliance

DoD contracts require DFARS (Defense Federal Acquisition Regulation Supplement) compliance — additional rules beyond standard FAR. Cybersecurity (CMMC), trafficking, country-of-origin, prohibited Chinese sources.

Core requirements

Requirement	Scope
DFARS 252.204-7012	Safeguarding covered defense info + cyber incident reporting
NIST SP 800-171	Underlying cybersecurity standard — 110 controls
CMMC 2.0 (Cybersecurity Maturity Model Certification)	Levels 1, 2, 3 — required for contracts handling FCI/CUI
DFARS 252.225-7012	Buy American Act — domestic content requirements for DoD
Section 889 NDAA	Prohibition on Chinese-origin telecom + video surveillance equipment
Anti-Trafficking (DFARS 252.222-7007)	Combat trafficking compliance plan
Country-of-origin (DFARS 252.225-7000)	Reporting
Counterfeit Parts (DFARS 252.246-7008)	Counterfeit detection + prevention

CMMC implementation pathway

- Level 1 (basic safeguarding)
- Self-assessment
- Level 2 (advanced)
- C3PAO (Certified Third-Party Assessor Organization) assessment
- Level 3 (expert)
- Government-led assessment

- Implementation: GAP analysis → SSP (System Security Plan) → POA&M; → C3PAO → Continuous Monitoring

Data sources

Source	Returns	Cost	Fallback
SAM.gov + Contract Opportunities	Primary federal opportunity portal	Free public	Direct sam.gov
USAspending.gov	Federal spending data + competitor analysis	Free public	Direct
GovWin IQ (Deltek)	Federal market intelligence + opportunity tracking	Subscription \$\$\$	GovTribe alternative
Bloomberg Government	Federal market intel + analytics	Subscription \$\$\$	Direct
govSpend (formerly SmartProcure)	State + local opportunity intel	Subscription	Direct
FedScoop + Federal News Network + Bloomberg Government	Industry intel	Free / sub	Direct
SBA dynamic small business search	Set-aside designation verification	Free public	Direct
CPARS — accessible via SAM Past Performance	Past performance database (limited public access)	Federal user only	Internal
Beta.SAM Vendor Profile + Capability Statement repository	Capability docs	Free public	Direct
Agency-specific portals (DoD, DOE, NASA, GSA, EPA, DHS, VA, USDA)	Agency-specific opportunities	Free public	Direct

Federal Capture Playbook

14 categories — each with research basis + named example. Compliance flags shown where applicable.

Category	Research basis	Named example
SAM.gov registration + UEI + capability statement	Annual SAM renewal + reps + certs + capability statement	Annual SAM.gov renewal + capability statement updated with 5 most-recent past performance projects + NAICS codes optimized + named PSC (Product Service Codes) — drives CO discoverability + 35% increase in unsolicited inquiries.

Category	Research basis	Named example
GSA Schedule (MAS) acquisition	GSA MAS proposal preparation + Schedule submission + named consultants	GSA MAS proposal via named consultant (FedSchedules, ezGSA, Schedule Consultants Inc.) — 9-15 mo timeline + \$40-120k cost — captures access to \$45B/yr in federal Schedule spend.
Set-aside certification stack maximization	SBA size standards + named cert programs	SBA + 8(a) + WOSB + HUBZone + SDVOSB stacking — drives 4 set-aside categories accessible + sole-source eligibility up to \$4.5-7M per contract — 6-12 mo to build stack + ~\$5k cost.
Mentor-Protégé Joint Venture (MPJV)	SBA Mentor-Protégé Program + named large firm partners	MPJV with named large prime (e.g. SAIC, Leidos, Booz Allen, CACI, ManTech, ICF, Deloitte Federal) — JV claims small-business status without aggregation — captures projects \$50-500M otherwise inaccessible.
CPARS rating optimization	CPARS narrative templates + CO + COR relationship discipline	Pre-award CO meeting + quarterly progress reports + named small business sub utilization + post-completion CPARS narrative review — moves rating from Satisfactory to Very Good over 3 contracts.
NIST 800-171 + CMMC 2.0 cyber compliance	Named C3PAO assessors + GAP analysis + named cyber tech	CMMC Level 2 prep with named C3PAO assessor (Redspin, Cybernance, Coalfire Federal, Schellman, Edwards Performance Solutions) — 9-15 mo prep + \$80-200k cost — qualifies for \$40M+ DoD work otherwise blocked.
Capture team + BD investment	Named federal BD professionals + capture process + Shipley methodology	Hire dedicated federal capture lead + Shipley capture process + named opportunity tracker — drives proposal win rate from 10% to 22% via earlier engagement + better-qualified pursuit.
Subcontractor + teaming network development	Named teaming partners + capability complementarity	Build teaming network of 8 complementary firms + active teaming agreements — captures 30% of work as prime + 30% as sub via reciprocal teaming.
Named-agency target list + relationship development	Named agencies + named CO/COR + spending patterns from USAspending	Target 5 named agencies (e.g. VA, GSA, DLA, Army, Air Force) + 25 named CO/COR + named small-business specialists — drives 60% of pipeline from 5 named relationships.
Past performance documentation + reusable proposal library	Named proposal library tools (Privia, Loopio, RFPIO) + standardized PP	Loopio or RFPIO proposal library + standardized past performance database — drops proposal time from 200 hrs to 80 hrs per submission + drives more submissions.

Category	Research basis	Named example
GSA Schedule + GWAC + MATOC pursuit (multi-vehicle)	Named federal contract vehicles + capacity-building strategy	GSA MAS + NASA SEWP + NIH NITAAC + USACE MATOC simultaneously — drives multiple capture paths + reduces single-vehicle dependence.
Federal FY-end (Sep 30) capture push	Federal FY-end use-or-lose dynamics + accelerated procurement	Q4 federal FY (Jul-Sep) capture push + accelerated proposals + named CO outreach for FY-end obligations — captures 40% of annual federal awards in 25% of calendar.
GSA OASIS / Polaris / Alliant 2 / similar GWAC pursuit	Named professional services GWACs	GSA OASIS+ pursuit + Alliant 2 pursuit + CIO-SP4 — captures access to \$50B+ in professional services + IT GWAC ceilings.
State + local government work as bridge	State + local opportunity portals + named state contract vehicles	State + local government work to build past performance + capability + bridge to federal — typical pathway: state work → state cooperative purchasing → federal Schedule → prime federal.

KPIs

- Federal pipeline value (total \$ identified + tracked)
- Pipeline coverage (pipeline / annual quota target)
- Win rate on submitted federal proposals (target > 18%)
- Federal revenue % of total revenue
- CPARS rating average (target Very Good or higher)
- Active federal contract vehicles (count: GSA + GWACs + agency)
- Set-aside certification count (active)
- CMMC certification level
- Mentor-Protégé / JV partnerships (active)
- Recompete win rate on federal contracts (target > 80%)

Cycle / calendar model

Window	Trigger	Action
Federal FY-end (Jul-Sep)	Use-or-lose obligations + accelerated procurement	Q4 federal FY capture max
Federal FY start (Oct)	New fiscal authority + new starts	Annual planning + new-pursuit pipeline
Annual: SAM renewal + reps + certs	Annual SAM.gov renewal cycle	Renewal discipline + cert update
Annual: GSA Schedule contract anniversary	Schedule renewal + MAS modifications	Schedule update + pricing review

Window	Trigger	Action
Annual: CPARS narrative review + dispute	CPARS performance period	Narrative update + contested rating dispute
Quarterly: pipeline review + capture board	Internal capture management	Pipeline qualification + bid/no-bid decisions
Annual: cyber + CMMC continuous monitoring	CMMC Level 2/3 ongoing requirements	Cyber posture + continuous monitoring
Multi-year: GWAC + IDIQ contract recompetes	5-10 yr GWAC cycles	Strategic recompile preparation

Issue library

- Federal-revenue concentration risk — universal — frame as commercial diversification + multi-vehicle
- Bid + proposal cost vs win rate — universal — frame as Shipley capture + better qualification
- CPARS rating disputes + narrative drift — universal — frame as ongoing CO/COR + post-period reviews
- Set-aside certification graduation — universal — frame as MPJV + capability transition + commercial pivot
- CMMC certification cost + complexity — universal — frame as named C3PAO + phased implementation
- Federal payment delays + cash flow — universal — frame as factoring + DCAA-compliant accounting
- GSA Schedule pricing pressure + IFF (Industrial Funding Fee) — universal — frame as pricing discipline + non-Schedule alternates
- Subcontractor teaming risk — universal — frame as written teaming + non-circumvention + IP protection
- Recompete loss risk — universal — frame as incumbent advantages + relationship + transition planning
- DCAA + cost accounting compliance — universal — frame as named DCAA-compliant accounting (Unanet, Deltek Costpoint, Procas, Jamis)

Report sections

Section	Contents
Federal contractor snapshot	NAICS + PSC + UEI + cert stack + active vehicles + revenue mix
Pipeline + capture board	Active opportunities + qualification + capture stage
Contract vehicle strategy	GSA + GWAC + MATOC + agency-specific vehicles
Set-aside cert + 8(a) roadmap	Active + pursuable certs + graduation planning
Top 5 federal capture moves	Drawn from federal_capture_playbook
CPARS + past performance plan	Rating trajectory + narrative discipline
CMMC + cyber compliance roadmap	Named C3PAO + cert level pursuit

Section	Contents
Teaming + JV + Mentor-Protégé pipeline	Named partners + active arrangements

Specificity rules

- Every named federal program must be real (SAM.gov, GSA MAS, NASA SEWP, NIH NITAAC, GSA OASIS+, GSA Alliant 2, NAVFAC MACC, USACE MATOC, CIO-SP4, FAR, DFARS)
- Every named cert must be real (SBA, 8(a), HUBZone, WOSB/EDWOSB, SDVOSB/VOSB, CMMC Level 1/2/3, NIST SP 800-171, Section 889, FAR Part 4)
- Every named C3PAO must be real (Redspin, Cybernance, Coalfire Federal, Schellman, Edwards Performance Solutions, A-LIGN — partial CMMC Marketplace list)
- Every named GSA consultant must be real (FedSchedules, ezGSA, Schedule Consultants Inc., RDB Group, GovDataHosting)
- Every named federal prime must be a verifiable currently-active prime (SAIC, Leidos, Booz Allen, CACI, ManTech, ICF, Deloitte Federal, Accenture Federal, Northrop Grumman, Raytheon BBN)
- Every named DCAA-compliant accounting platform must be real (Unanet, Deltek Costpoint, Procas, Jamis, Jamis Prime, FedRamps)
- Every named federal BD intelligence platform must be real (GovWin IQ by Deltek, Bloomberg Government, govSpend, FedScoop, GovTribe)
- Compliance language must align with current FAR + DFARS at the time of writing

BATCH 7 — CLOSING SUMMARY

Score impact + integration

Batch 7 closes the four worst spec gaps from the prior rating. After Batch 7 every sub-cluster scores ≥ 80 from spec depth alone. The remaining gap to 95+ across all sub-clusters comes from Batches 8-12 (verified case studies, raw failure data, classifier expansion, sector polish, final integration).

Score impact

Sub-cluster	Before Batch 7	After Batch 7	Lift
RIA / Wealth Management (52.C)	68/100	92/100	+24
Civil / Heavy Construction (23.C)	55/100	90/100	+35
Pharma / Medical Device (subset of 31-33.C)	~75/100	94/100	+19
Federal Contractor (any sector with $\geq 15\%$ federal)	varies	+10-15 modifier lift	+10-15

Engineering integration

The Master Router needs three additions to handle specialty templates correctly:

1. Profile registry gains a 'template_type' field. Values: 'standard' (the 86 profiles in Batches 1-6), 'specialty_replacement' (RIA, Civil/Heavy, Pharma — replace base profile), 'specialty_modifier' (Federal Capture — applied alongside base profile).
2. Specialty replacement templates are routed by NAICS-6 + sub-cluster classification:
523930/523991/523999 → RIA template (replaces finance.ria_wealth_management);
237310/237120/237130 → Civil/Heavy template (replaces construction.civil_heavy); pharma + medical device subset of 325xxx + 339xxx → Pharma template.
3. Specialty modifier templates are routed by an additional input flag (federal_revenue_pct). When this flag is $\geq 15\%$, the modifier template is applied alongside the base profile — final report has BOTH playbook sections.
4. Report renderer gains conditional rendering logic that knows how to display compliance_filter, bid_pipeline_sources, fda_submission_lifecycle, sam_gov_pipeline, etc. — this is the specialty_renderer.py pattern from this batch.
5. Compliance filter (RIA template only) is enforced as a hard rule: the Master Router refuses to render RED-flag recommendations without an explicit override flag from the user (e.g. 'I have outside counsel review available'). BLOCKED items are never rendered.

What still needs Batches 8-12 to reach universal 95/100

Batch 8 — Real TAM (Census CBP) + raw BLS BED failure rates per NAICS-4. Batch 9 — Verified case study numbers replacing illustrative claims. Batch 10 — Classifier keyword expansion (~45,000 phrase variations). Batch 11 — Sector polish (deepen 11/21/22, split Education + Information further). Batch 12 — Final integration + golden tests + verification scripts.

Total spec status after Batch 7

Batch	Coverage	Pages
1-2	Architecture + Sectors 11/21/22/44-45/62/71/72/81	114
3	Sectors 23/52/53/54	69
4	Sectors 31-33/42/48-49/56	62
5	Sectors 51/61 + Out-of-Scope + Failure DB + Roadmap + Appendices	77
6	Master Routing Table — 1,466 entries → 90 profiles	71
7	Specialty Templates (RIA + Civil/Heavy + Pharma + Federal)	this batch
Total so far	—	393 + this batch